Regular Study Days, Not Click Volume: Temporally Ordered Evidence from Ebook Reading Traces and Course-Aligned Mathematics Outcomes

Abstract

Fine-grained ebook logs are often interpreted as indicators of engagement, but high event volume alone is a weak basis for learning analytics claims. This study asks whether temporally organized same-course ebook activity predicts later assessment performance in K-12 mathematics, and whether such associations survive stricter observational controls. We linked Bookroll/xAPI traces with grade/test records and Learning Management System (LMS) course metadata, mapping events to a common student-course-month feature layer. The analysis focused on course-embedded junior-high mathematics assessments where same-course ebook traces met explicit coverage criteria. Models used normalized assessment scores, student fixed effects, and assessment-occasion fixed effects, comparing each student against their own other assessments while controlling for course/test/date difficulty. A stricter robustness model added student-course fixed effects to remove stable student-by-course differences.

Across course-embedded regular exams and unit/chapter tests, active study days before the assessment were consistently associated with higher outcomes after adjusting for raw event volume and behavior composition. The association was stable across 3-, 6-, and 12-month windows, remained positive after adding student-course fixed effects, was strongest for lower junior-high mathematics, and was absent for external Benesse tests. Temporal strategy models showed that category-level strategy advantages were largely explained by measured regularity, especially active days. A future-activity placebo check strengthened temporal interpretation: activity after the assessment did not reproduce the positive pre-test

association. The results support the interpretation that regular same-course ebook engagement is a stronger indicator of course-aligned learning outcomes than click volume or annotation-heavy activity alone.

1. Introduction

Digital textbooks and ebook readers can record detailed traces of learners' interactions with course materials. These traces are attractive for learning analytics because they are collected during ordinary study rather than in artificial laboratory tasks. However, trace data also create a familiar risk: many learning analytics systems may be optimizing the wrong signal. A student who generates many clicks is not necessarily studying effectively, and an aggregate engagement count may mix navigation, repeated opening, annotation, review, and post-assessment remediation.

This paper takes a narrower and more defensible approach. Instead of asking whether students who generally use ebooks more perform better, we ask whether same-course ebook activity before course-aligned assessments is associated with better outcomes within students and within assessment occasions. This distinction matters. Between-student comparisons are vulnerable to stable confounding: higher-achieving or more motivated students may both study more and score higher. Assessment-level comparisons are also confounded because some tests are harder, some courses are structured differently, and some subject-grade combinations use ebooks more heavily. We therefore use student fixed effects and assessment-occasion fixed effects to ask whether a student's own variation in pre-test ebook regularity aligns with their own variation in outcomes across assessments.

The empirical setting is a K-12 ebook learning environment in Japan. The data include grade/test outcomes, Learning Management System (LMS) course metadata, and Bookroll/xAPI traces. A methodological contribution is the construction of a same-course longitudinal trace layer that links ebook activity to the specific course and assessment context in which learning outcomes are observed. Event records are summarized at a student-course-month grain and joined to cleaned score outcomes only after validating course linkage and temporal ordering.

The analysis leads to a clear result. The strongest and most stable predictor is not raw event volume. It is the number of active same-course study days before course-embedded assessments. The mechanism is specific: it appears for regular school exams and unit/chapter tests, but not for external Benesse tests. This assessment-family contrast is important because it prevents an overgeneralized claim. Regular same-course ebook activity appears related to course-aligned outcomes, not necessarily to broader standardized achievement.

The paper makes three contributions. First, it provides a same-course trace construction approach that links ebook behavior to the specific course and assessment context in which outcomes are observed. Second, it demonstrates that regularity of same-course study days is more robust than raw click volume for predicting course-aligned outcomes. Third, it uses a causal-cautious observational design with student fixed effects, stricter student-course fixed effects, assessment fixed effects, assessment-family contrasts, and a future-activity placebo check. Together, these steps move the analysis beyond descriptive trace mining toward a broader validity principle: trace data become more meaningful when they are course-aligned, temporally ordered, and interpreted against the assessment context they are meant to inform.

1.1 Related Work and Positioning

Learning analytics research has long relied on behavioral traces from digital learning environments, yet the validity of such traces remains a central challenge (Siemens & Long, 2011; Winne, 2020; Wise & Shaffer, 2015). Log data are precise records of interaction, but they are not automatically precise records of learning. Page turns, clicks, dwell time, highlights, and annotations can reflect purposeful study, confusion, interface friction, repeated opening, or post-assessment review. This ambiguity is especially important in ebook environments, where learners may interact with the same material in many ways and for many reasons. The present study addresses this validity problem by asking whether ebook behavior remains predictive after three restrictions are imposed: events must be linked to the same course as the assessment, behavior must occur before the assessment, and estimates must compare students within the same assessment context.

Research on self-regulated learning and distributed practice provides a theoretical basis for focusing on regularity rather than volume (Cepeda et al., 2006; Dunlosky et al., 2013). Distributed engagement across multiple study occasions is more consistent with sustained preparation than a large number of actions concentrated in a short period. The active-days measure used here is not a direct psychological measure of self-regulation, motivation, or metacognition. It is a behavioral indicator of repeated same-course study occasions. This distinction is important: the paper does not claim to infer students' internal states from logs. Instead, it tests whether an observable temporal pattern of engagement is associated with course-aligned achievement.

Digital textbook and ebook analytics studies often examine page views, navigation, bookmarks, highlights, memos, and other interaction types as indicators of engagement or strategy (Ogata et al., 2015). A limitation of this approach is that individual action types can be difficult to interpret in isolation. More annotations may indicate deeper processing, but they may also indicate confusion, copying, or task requirements. More navigation may indicate purposeful review, but it may also indicate searching or disorientation. The contribution of the present analysis is therefore not another ranking of raw action types. It shows that annotation-heavy and click-heavy behavior are not sufficient explanations once regularity and assessment context are considered. Regular same-course active study days provide a more stable signal than raw event volume.

Methodologically, this study contributes a causal-cautious observational design for ebook trace analysis. Student fixed effects reduce confounding from stable learner characteristics such as prior achievement, general motivation, and background. Assessment-occasion fixed effects control for course, test name, test date, and shared assessment difficulty. Student-course fixed effects add a stricter robustness check by absorbing persistent learner-course differences. Family-specific models distinguish course-embedded assessments from external standardized tests, and a future-activity placebo checks whether activity after the assessment reproduces the pre-assessment association. These choices connect the learning analytics concern with trace validity to the learning-design concern with alignment: behavioral traces should be interpreted only in relation to the instructional context and assessment purpose they plausibly represent (Gašević et al., 2015; Lockyer et al., 2013). They do not turn observational data into experimental evidence, but they substantially raise the evidentiary standard beyond cross-sectional prediction or descriptive dashboard analytics (Angrist & Pischke, 2009; Shadish et al., 2002).

2. Research Questions

RQ1. Within students and assessment occasions, is pre-test same-course ebook activity associated with mathematics assessment outcomes?

RQ2. Is the association better explained by raw event volume, behavior composition, or regularity of active study days?

RQ3. Does the active-days mechanism generalize across assessment families, or is it specific to course-embedded assessments?

RQ4. Does a future-activity placebo check weaken or strengthen the temporal interpretation?

3. Data and Linkage

The source data combine three kinds of records: assessment outcomes, LMS course metadata used to recover grade/subject context, and Bookroll/xAPI event traces. Rows with missing test conduct dates were excluded from test-window analysis because the temporal ordering of behavior and outcome could not be established. The source scaled score field was not used because it was zero in dated score rows. The primary outcome is therefore a normalized score:

$$\text{normalized score} = \frac{\text{quiz} - \text{minimum score}}{\text{maximum score} - \text{minimum score}}.$$

The learning environment is part of LEAF, the Learning and Evidence Analytics Framework, which connects learning tools and learning record infrastructure to support data-informed educational practice in Japanese settings (Ogata et al., 2022, 2024). Within this environment, Bookroll is the ebook reader used for course materials. Students access assigned digital texts through Bookroll, and interactions such as opening and closing content, page navigation, markers, memos, search, and embedded quiz actions are logged as xAPI statements. This makes Bookroll traces suitable for studying when and how students return to same-course materials before assessments, while LEAF provides the broader infrastructure for connecting those traces to course context and outcomes.

The ebook trace layer required careful preprocessing because logs were collected over multiple school years and courses. The pipeline summarizes trace records at a common feature grain: student by course by event month. For older Bookroll records, course context was often not reliable in the xAPI context field, so events were linked through content identifiers, Bookroll content-directory ownership, and the LMS/score course identifier. Only uniquely mapped content-course links were retained; ambiguous and unmapped contents were excluded from same-course features. For newer xAPI records, non-empty context identifiers were used directly as course identifiers. In the current modeled strong cells, however, the 3-month same-course evidence comes from the older mapped source; new-source rows are zero because dated score outcomes end before the new xAPI format contributes materially.

The old-source bridge produced 33,337 content-directory-owner candidate rows. Of these, 25,943 were high-confidence candidates, 6,356 contents mapped uniquely to one score-course identifier, and 19,331 ambiguous candidate rows were excluded. In event-volume terms, 15,243,211 old events and 4,117,740 new events were represented in same-course mapped features for score students, while 59,209,738 old events were skipped because the content could not be uniquely mapped to a course. These exclusions are deliberate: they reduce coverage but protect the same-course claim from weak linkage.

Table 1 summarizes the sample construction. A strong grade/subject/test-family cell is defined before fixed-effect modeling as one with at least 100 valid outcomes, at least 100 students, at least 100 score rows with same-course xAPI activity in the 3-month pre-test window, and at least 50% 3-month same-course xAPI coverage, where coverage is calculated over all clean score rows in the cell. This rule prevents large but weakly linked cells from being treated as strong evidence.

Stage	Records	Role in analysis
Raw score records	67,672	Initial assessment-score table
Dated score records	43,180	Rows with test date available for

Stage	Records	Role in analysis
		temporal ordering
Valid normalized outcomes	42,548	Rows with usable normalized score
Strong-cell valid rows	14,248	Rows in 12 cells meeting the strong-cell rule
Course-embedded rows	9,906	Regular exams and unit/chapter tests
External Benesse contrast rows	3,636	External assessment comparison
Break-after-test rows	706	Retained in diagnostics but not the main course-embedded claim
Global fixed-effect rows	14,155	Strong-cell rows after student and assessment FE filtering
Course-embedded student+assessment FE rows	9,781	Main course-aligned fixed-effect comparison
Course-embedded student-course+assessment FE rows	9,547	Stricter robustness comparison
Same-course pre-test ebook events in strong cells	8,553,155	Trace richness in the 3-month pre-test window

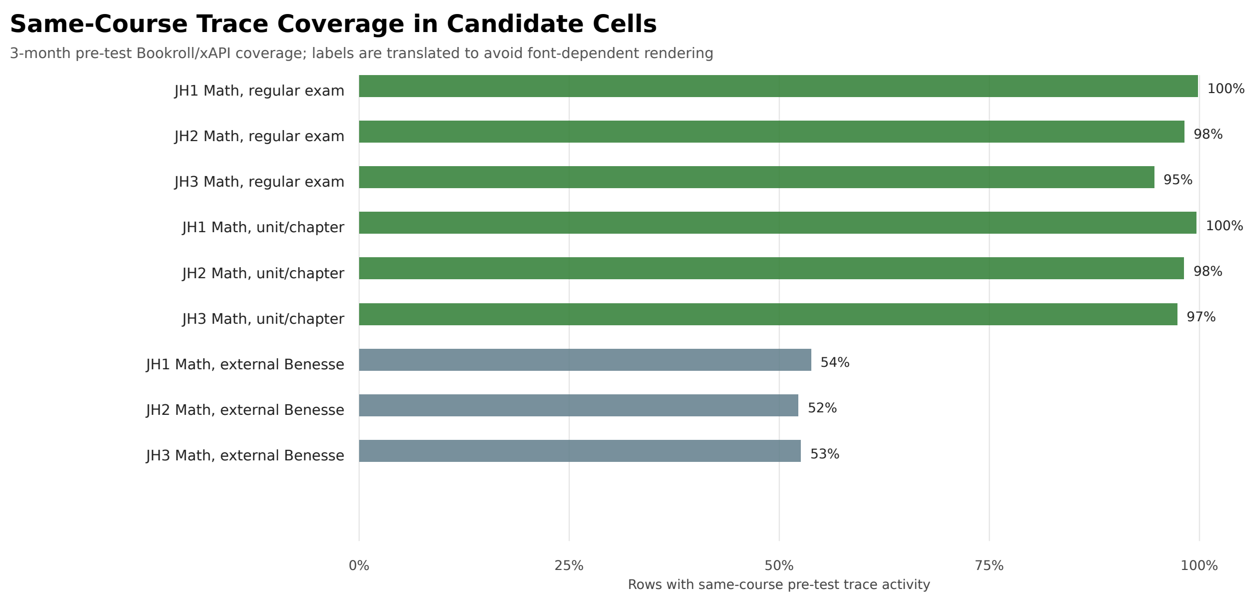

Figure 1. Same-course pre-test xAPI coverage in candidate analysis cells. The modeled junior-high mathematics cells have near-complete 3-month same-course trace coverage, supporting their use as the primary evidence base.

For the main models, we focus on the strong course-embedded junior-high mathematics cells: regular school exams and unit/chapter tests. External Benesse tests are analyzed separately as a theoretically important contrast because they are more formal and less directly tied to a single course's ebook materials.

3.1 Ethics and Data Governance

The analysis uses de-identified institutional learning records for secondary analysis. Student-level records remain in local analysis files and are not exported in the manuscript, figures, tables, or agent-facing summaries. The paper reports only aggregate counts, coefficients, confidence intervals, and derived diagnostics. Any submission version should state the applicable institutional approval, administrative permission, or consent basis for the dataset. The analysis pipeline was designed to minimize exposure of K-12 student data: raw and row-level files remain local, linkage uncertainty is documented, and ambiguous content-course links are excluded rather than manually inferred.

4. Measures

The outcome is normalized assessment score. The main behavior feature is log_active_days, defined as log(1 + active days), where active days is the count of distinct days with same-course ebook activity in the pre-test window. Other behavior features include log event volume, defined as log(1 + events), and navigation, memo, marker, and content-session shares, each divided by max(events, 1). Zero-activity rows are retained: log features equal 0 and event-share features equal 0. These features are computed for 3-, 6-, and 12-month windows before the assessment. For temporal strategy analysis, the 3-month pre-test period is divided into early, middle, and late phases.

Time-spent features were considered but not used as the primary mechanism in the primary analysis. The logs include elapsed time since the previous activity, but such values can overstate study time when a learner leaves a page open, switches tasks, or resumes after an idle period. Without a carefully validated timeout rule, time spent risks adding a noisier proxy than active days. For this paper, active days provide a more conservative temporal measure. Time-spent variants should be reported as a robustness analysis after defining and validating session-time caps.

Temporal strategy categories include distributed navigation, distributed sustained activity, late-intensive activity, early-declining activity, intermittent activity, single-month activity, and no same-course activity. These categories are descriptive and are not treated as causal mechanisms by themselves. The stricter model includes both strategy categories and behavior features to test whether strategy labels retain explanatory value after measured behavior regularity and composition are included.

5. Analysis Design

The analysis is observational and causal-cautious. We do not claim random assignment to ebook behavior. The strongest models use two-way fixed effects:

$$y_{iat} = \alpha_i + \gamma_a + \beta x_{iat} + \epsilon_{iat}.$$

where y_iat is the normalized score for student i on assessment occasion a, alpha_i is a student fixed effect, and gamma_a is an assessment-occasion fixed effect defined by course, test name, and test date. This design compares a student with themselves across assessment occasions while controlling for difficulty and context shared by students taking the same assessment.

The design reduces confounding from stable student ability, stable motivation, background, and fixed differences across assessments. It does not remove time-varying confounding such as changes in effort, teacher support, offline study, or exam-specific preparation. For this reason, the paper uses terms such as associated with, consistent with, and supports a causal-cautious interpretation.

Assessment occasions with fewer than 20 rows are removed before fixed-effect estimation. Student fixed-effect models require at least two rows per student and within-student variation in at least one predictor. Student-course models apply the same logic at the student-course level. Reported coefficients are standardized residual associations unless explicitly labeled as unstandardized. Confidence intervals for the global window model use student-cluster bootstrap resampling; family-specific, student-course, and placebo models use student-clustered standard errors. Standard errors are clustered by student because repeated observations are nested within learners across courses and assessment occasions.

We add four robustness checks. First, we repeat models across 3-, 6-, and 12-month windows. Second, we estimate family-specific models for regular school exams, unit/chapter tests, and Benesse tests. Third, we estimate stricter models with student-course fixed effects:

$$y_{icat} = \alpha_{ic} + \gamma_a + \beta x_{icat} + \epsilon_{icat}.$$

where alpha_ic absorbs stable differences for student i in course c. This check asks whether the result remains after controlling for a student's persistent course-specific strength, weakness, or engagement level. Fourth, we use a future-activity placebo: if activity after an assessment predicts the previous score in the same positive direction, the temporal interpretation is weaker.

6. Results

6.1 Regularity Outperforms Raw Event Volume

The fixed-effect models show a consistent pattern. Active days remain positive across pre-test windows, while raw event volume does not. In the broader strong-cell model, active days are positively associated with outcomes, and this pattern grows slightly in 6- and 12-month windows. Raw event volume is weak and trends negative after adjusting for active days and behavior composition. This suggests that regularity of engagement, not click quantity, is the meaningful signal. Because the broader model includes external Benesse and break-after-test rows, it is used as a general robustness check; the primary substantive claim rests on the course-embedded regular-exam and unit/chapter-test models.

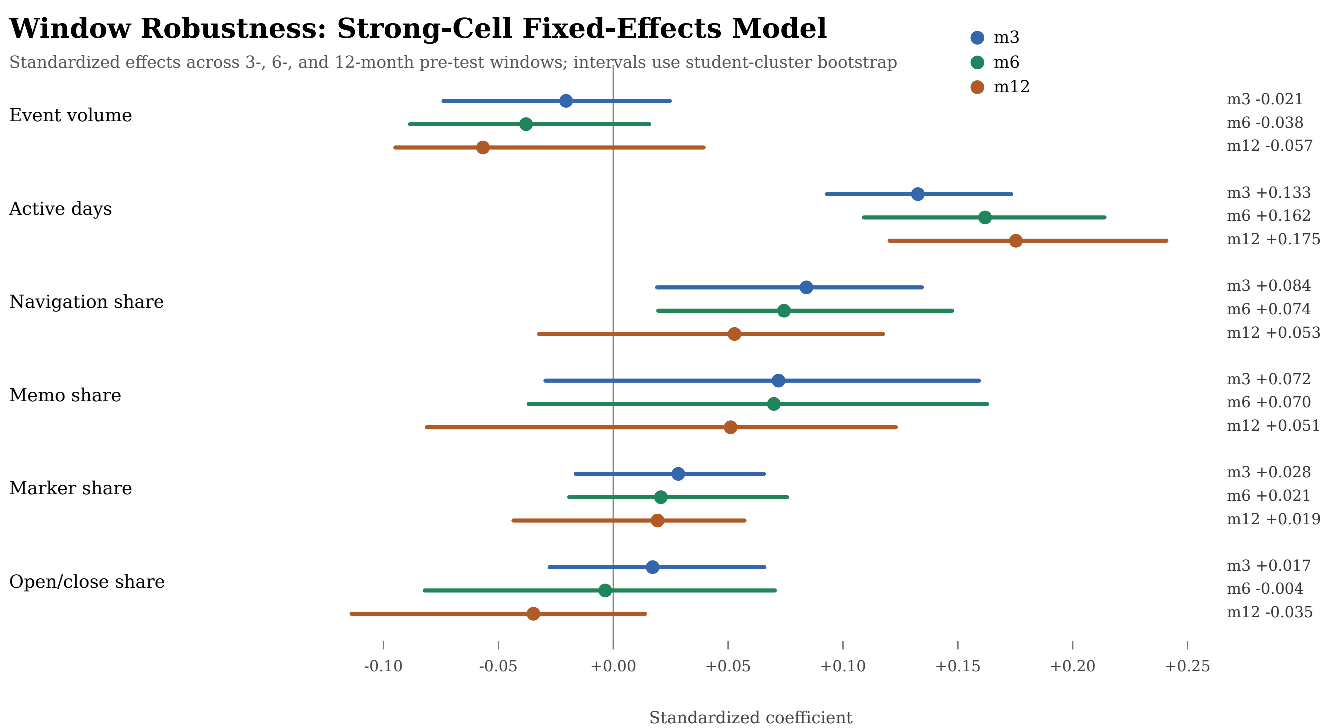

Figure 2. Window robustness of behavior effects. Active days remain positive across pre-test windows, whereas event volume does not provide a stable positive signal.

6.2 Temporal Strategy Effects Are Largely Explained by Measured Regularity

Temporal strategy categories are positive relative to no same-course activity when entered alone. However, when behavior features are added, strategy coefficients shrink and confidence intervals cross zero. Active days remain positive in the adjusted model. This means the apparent advantage of named strategies is largely explained by measured regularity of engagement. The total strategy model uses 14,097 observations from 933 students and 370 assessment occasions; the adjusted model uses 14,101 observations from 935 students and 370 assessment occasions.

Table 2. Strategy contrasts before and after behavior adjustment.

Temporal strategy	Total beta	Total 95% CI	Adjusted beta	Adjusted 95% CI
distributed_navigation	+0.068	[+0.024, +0.111]	+0.043	[-0.044, +0.130]
distributed_sustained	+0.095	[+0.035, +0.154]	+0.061	[-0.064, +0.187]
late_intensive	+0.100	[+0.032, +0.167]	+0.066	[-0.082, +0.213]
intermittent_activity	+0.141	[+0.047, +0.234]	+0.091	[-0.112, +0.293]
early_declining	+0.072	[+0.014, +0.129]	+0.048	[-0.074, +0.171]
single_month_activity	+0.048	[+0.010, +0.087]	+0.042	[-0.036, +0.120]

6.3 The Active-Days Mechanism Is Specific to Course-Embedded Assessments

The family-specific models show that active days are robustly positive for regular school exams and unit/chapter tests. The same mechanism does not appear for external Benesse tests.

Table 3. Active-days effects by assessment family.

Assessment family	Window	Active-days beta	95% CI	p	Rows	Students
school_regular_exam	m3	+0.056	[+0.015, +0.097]	0.007	5,776	475
school_regular_exam	m6	+0.068	[+0.032, +0.105]	0.000	5,776	475
school_regular_exam	m12	+0.070	[+0.034, +0.106]	0.000	5,769	474
unit_or_chapter_test	m3	+0.077	[+0.033, +0.120]	0.001	3,961	457
unit_or_chapter_test	m6	+0.086	[+0.044, +0.127]	0.000	3,961	457
unit_or_chapter_test	m12	+0.091	[+0.050, +0.132]	0.000	3,961	457
external_benesse	m3	-0.046	[-0.108, +0.015]	0.140	3,484	932
external_benesse	m6	-0.033	[-0.097, +0.030]	0.303	3,484	932
external_benesse	m12	-0.033	[-0.097, +0.030]	0.303	3,484	932

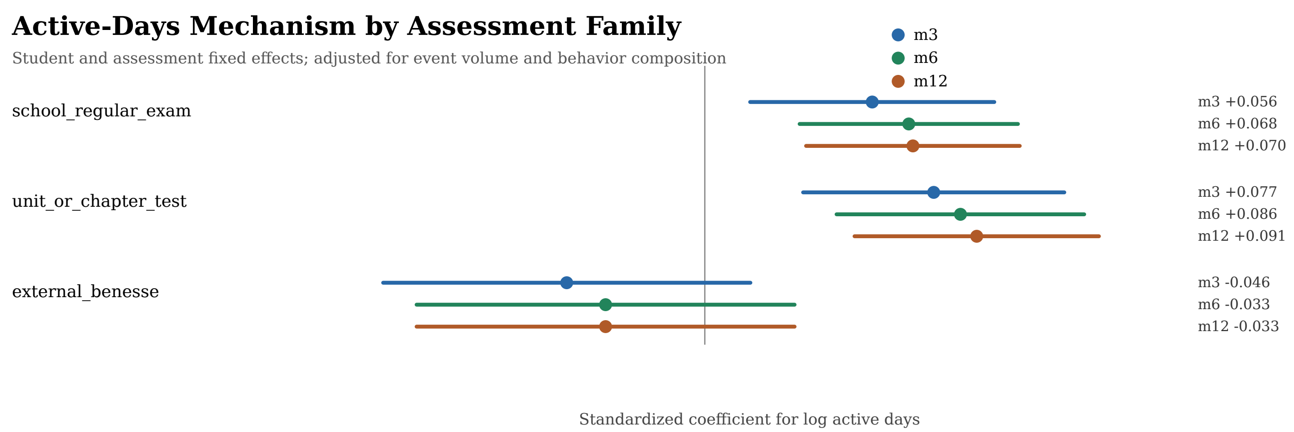

Figure 3. Active-days mechanism by assessment family. The positive association appears in course-embedded assessments and not in the external Benesse contrast.

This contrast is one of the most important findings. The Benesse contrast is consistent with a course-alignment interpretation, although differences in test purpose, content coverage, stakes, timing, and preparation culture may also contribute. The result should therefore be read as evidence against a generic achievement-signal interpretation, not as proof that content alignment is the only explanation.

6.4 The Result Survives Student-Course Fixed Effects

The stricter robustness model replaces student fixed effects with student-course fixed effects while retaining assessment-occasion fixed effects. This is a materially stronger comparison: it asks whether a student's deviations from their own usual pattern within the same course predict assessment deviations, not merely whether the same student is more active in some courses or periods than others. The active-days coefficient remains positive for course-embedded assessments across all windows. In the 3-month window, the student-course fixed-effect estimate is +0.056 for course-embedded assessments, +0.052 for regular exams, and +0.068 for unit/chapter tests. The external Benesse contrast remains non-positive.

Table 4. Student-course fixed-effect robustness for active days.

Scope	Window	Student FE beta	Student-course FE beta	Student-course 95% CI	p	Rows	Stude cours
Course-embedded	m3	+0.062	+0.056	[+0.021, +0.091]	0.002	9,547	711
Course-embedded	m6	+0.074	+0.067	[+0.038, +0.096]	<0.001	9,547	711
Course-embedded	m12	+0.078	+0.079	[+0.050, +0.108]	<0.001	9,540	710
Regular exams	m3	+0.056	+0.052	[+0.007, +0.097]	0.023	5,540	710
Unit/chapter tests	m3	+0.077	+0.068	[+0.025, +0.111]	0.002	3,952	690
External Benesse	m3	-0.046	-0.034	[-0.104, +0.036]	0.342	3,020	1,510

These estimates strengthen the causal-cautious interpretation. They do not prove that reading activity caused the score increase, but they make several weaker explanations less plausible: stable student ability, stable course preference, and persistent student-course differences are all absorbed by the fixed effects.

For practical interpretation, the unstandardized student-course fixed-effect estimate for the course-embedded 3-month model is 0.0175 normalized-score points per one-unit increase in log(1 + active days). Among active rows in this model, the first quartile, median, and third quartile are 5, 9, and 14 active same-course study days. Moving from 5 to 14 active days corresponds to approximately 1.60 normalized-score percentage points, holding student-course and assessment occasion fixed. Moving from 3 to 10 active days corresponds to approximately 1.77 percentage points. These are modest but meaningful within-assessment differences, especially because they are estimated after removing stable student-course and shared assessment differences.

Table 5. Practical effect-size and sensitivity diagnostics.

Quantity	Value
Student-course FE beta for log(1 + active days)	0.0175 normalized-score points
Clustered SE	0.0056
Active days among active rows, Q1 / median / Q3	5 / 9 / 14 days
Predicted difference from Q1 to Q3 active days	1.60 percentage points
Predicted difference from 3 to 10 active days	1.77 percentage points
Excluding top 1% event-volume rows	beta = 0.0176, SE = 0.0056
Coverage threshold sensitivity, 40%-90%	beta = 0.0175 across thresholds
Active-day bin check	positive for all 3+ day bins; largest for 21+ days
Residual correlation: log events vs log active days	0.480

6.5 The Mechanism Is Strongest in Lower Junior-High Mathematics

Grade/subject consistency checks show that the active-days coefficient is positive in nearly all estimated junior-high mathematics cells, with the clearest effects in first- and second-year junior-high mathematics. Third-year mathematics is mostly positive but less precise. This pattern suggests that the result is not driven by a single grade, while also showing that the mechanism is not uniform across all grade levels.

Supplementary Table S1 gives the compact grade-level breakdown.

6.6 Future Activity Does Not Reproduce the Positive Pre-Test Effect

The future-activity placebo supports temporal interpretation. For regular exams and unit/chapter tests, pre-test active days are positive. Future active days after the assessment are not positive; for unit/chapter tests, the future coefficient is negative. This does not prove causality, but it weakens the alternative explanation that the active-days result is simply a stable marker of generally motivated students.

Table 6. Future-activity placebo check.

Assessment family	Window	Active-days beta	95% CI	p	Rows	St
school_regular_exam	pre_m3	+0.056	[+0.015, +0.097]	0.007	5,776	47
school_regular_exam	future_m3_placebo	-0.037	[-0.086, +0.011]	0.128	5,773	47
unit_or_chapter_test	pre_m3	+0.077	[+0.033, +0.120]	0.001	3,961	45
unit_or_chapter_test	future_m3_placebo	-0.061	[-0.115, -0.007]	0.028	3,999	47

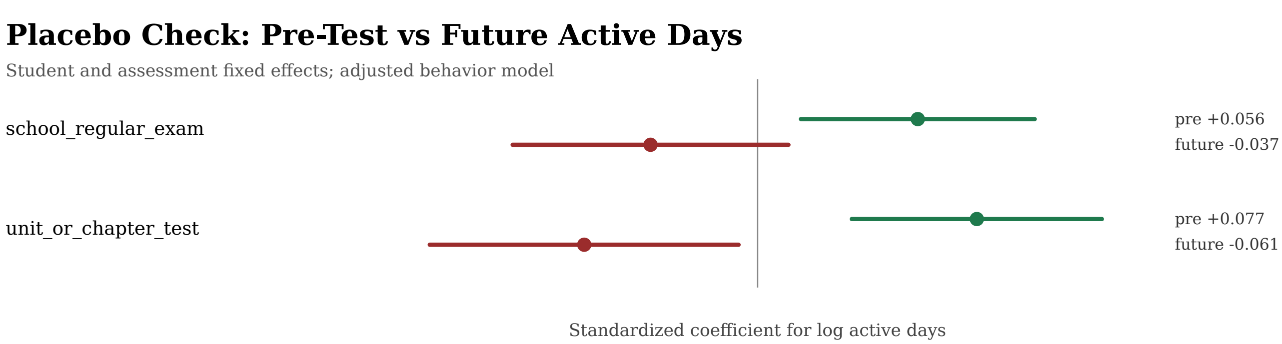

Figure 4. Pre-test versus future-placebo active days. Future activity does not reproduce the positive pre-test association.

The negative future coefficient for unit/chapter tests should be interpreted cautiously. It may reflect remediation or continued study after weaker performance rather than an inverse causal effect of future study.

7. Discussion

The central finding is that regular same-course ebook activity predicts course-aligned mathematics outcomes more reliably than raw event volume. This distinction is important for both theory and practice. From a theoretical perspective, the result suggests that temporal regularity is closer to a meaningful learning process than generic engagement volume. From a practical perspective, it warns against dashboards or interventions that reward more clicks without considering whether the activity is distributed across study days and aligned with course assessments. The design principle is not simply "use active days instead of clicks"; it is that trace validity depends jointly on temporal ordering, course alignment, and assessment context.

The assessment-family contrast sharpens the contribution. The mechanism holds for regular exams and unit/chapter tests, but not for Benesse external tests. This suggests that the observed behavior-outcome link is not a generic achievement signal. It is more plausibly tied to course-aligned preparation and interaction with materials relevant to the upcoming assessment.

The student-course fixed-effect results raise the evidentiary standard further. A stable student-by-course explanation is no longer sufficient: the association remains when the model compares a learner against their own other assessments within the same course context. This makes regularity a more credible behavioral signal than raw event volume and a more defensible basis for learning analytics intervention design.

The temporal and strategy analyses also prevent overclaiming. Strategy labels alone are not enough. Once active days and behavior composition are added to the model, category effects shrink. The interpretation should therefore focus on regular study days as the measurable mechanism, with strategy categories used as descriptive typology rather than causal constructs.

The practical implication should be cautious rather than punitive. Dashboards should not reward high click volume, but neither should they penalize low trace activity automatically. Offline study, peer study, paper-based preparation, and activity in other tools remain unobserved. A mature learning analytics system should identify regular, course-aligned engagement before assessment as a useful signal for reflection and support while avoiding deficit labels for students whose learning activity is not fully captured by Bookroll.

8. Limitations

First, this is an observational study. Student, student-course, and assessment fixed effects reduce important confounding, but time-varying unobserved factors remain. Second, same-course trace linkage requires reliable course mapping. Ambiguous content-course links were excluded, but linkage uncertainty should still be acknowledged. Third, the strongest evidence comes from junior-high mathematics, especially lower grades. The findings should not be generalized to all subjects or assessment types without additional validation. Fourth, offline study is unobserved. A student may study on paper, with peers, or in other systems, and those activities may interact with ebook behavior. Fifth, the current strong-cell evidence is dominated by older mapped Bookroll records; future work should re-test the mechanism when more post-2025 outcomes are available under the newer xAPI format.

9. Conclusion

This study shows that fine-grained ebook traces can support stronger learning analytics claims when they are linked to course context, temporally ordered, and modeled with causal-cautious controls. The main result is not that more clicks predict better outcomes. Rather, within students, assessment occasions, and stricter student-course comparisons, more regular same-course active study days before course-embedded mathematics assessments are associated with higher outcomes. The association is stable across windows, strongest for course-aligned assessments, absent for external Benesse tests, and not reproduced by future activity after the assessment.

The field-level implication is straightforward: learning analytics systems should not optimize for more clicks when the more defensible signal is regular, course-aligned study before assessment. The broader contribution is a validity principle for trace-based analytics: the meaning of a behavioral trace depends not only on what was logged, but also on when it occurred, which course it belonged to, and which assessment context it is used to explain.

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